

Jerry Felix

Backend Developer

Email | +91 93453 64086 | Thanjavur, Tamil Nadu
LinkedIn | GitHub | Codeforces | portfolio

About Me

Backend Developer with 2.7+ years of experience designing, developing, and optimizing scalable backend applications using ASP.NET Core, C#, Golang, PostgreSQL, ClickHouse, Elasticsearch, and Kafka. Experienced in building RESTful APIs, microservices, distributed systems, event-driven architectures, and AI-powered applications leveraging LLMs, RAG, LangChain, LangGraph, CrewAI, and vector databases. Strong expertise in backend optimization, database performance tuning, system design, security implementation, and enterprise software development.

Work Experience

Sardonyx Technologies Pvt. Ltd Jun 2024 – May 2026
Backend Developer

- Designed and implemented microservice-based backend architectures following SOLID principles and clean architecture practices.
- Integrated PostgreSQL, ClickHouse, and Elasticsearch to support high-volume transactional and analytical workloads.
- Developed RBAC-based authentication and authorization mechanisms for enterprise security applications.
- Built Kafka-based asynchronous processing pipelines to improve scalability and system reliability.
- Participated in Agile development processes including sprint planning, code reviews, testing, and production deployments.
- Optimized SQL queries, indexing strategies, and backend services to reduce latency and improve throughput.

Sardonyx Technologies Pvt. Ltd Oct 2023 – Apr 2024
Trainee Backend Developer

- Completed a 6-month intensive technical training program focused on real-time backend development.
- Built strong foundations in .NET, REST API design, SQL optimization and debugging.

Skills

ASP.NET Core

Entity Framework Core

React

Next

C#

Python

Golang

Database Design

PostgreSQL

ClickHouse

Qdrant

Elasticsearch

Kafka

LangChain

LangGraph

CrewAI

AI Agents

RAG

Vector Database

LLM

Docker

Git/GitHub

CI/CD

Vercel

API Development

Microservices

SDLC

System Design

Distributed Systems

Projects

EBuy Ecommerce Application

Live - <https://ebuy-fawn.vercel.app/>

- Built a responsive ecommerce web application using React.
- Implemented reusable UI components and optimized client-side performance.
- Automated deployment using GitHub and Vercel CI/CD pipelines.

AI Developer Agent

Status: In Development

- Building an AI-powered coding assistant using Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG).
- Implementing semantic search using embeddings and Qdrant vector databases.
- Developing tool-calling workflows and multi-agent orchestration using LangGraph and CrewAI.
- Designing scalable backend services for automated code generation and task execution.

AI Personal Assistant Agent

Status: In Development

- Developing an intelligent AI assistant capable of reasoning, task execution, and contextual decision-making.
- Implementing long-term memory systems using vector databases and retrieval mechanisms.
- Integrating workflow automation, tool-calling, and adaptive execution pipelines.
- Leveraging LLMs for natural language understanding and conversational interactions.

Achievements

- Optimized PostgreSQL query performance and introduced ClickHouse for analytical workloads, improving large-scale data processing efficiency.
- Designed and implemented Elasticsearch-based search capabilities for querying large volumes of structured and unstructured data.
- Refactored backend services through technology migration and data pipeline implementation, enhancing scalability and maintainability.
- Implemented RBAC, structured logging, SAST and DAST security practices for enterprise applications.

Certifications

Microsoft Azure Fundamentals (AZ-900) – In Progress

Microsoft Azure AI Fundamentals (AI-900) – In Progress

Cyber Security & Ethical Hacking for Beginners – HCL GUVI (2023)

Education

Bachelor of Engineering (Computer Science and Engineering)

Parisutham Institute of Technology and Science

Graduated: 2023

Language Skills

- English – Proficient
- Tamil – Native Proficiency